6 $\mathbf{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ $\mathbf{b} = \begin{pmatrix} 5 \\ -2 \end{pmatrix}$

Find $2\mathbf{a} - 3\mathbf{b}$ as a column vector.

$2a = (6, 8)$, while $3b = (15, -6)$.

$2a - 3b = (6 - 15, 8 - (-6)) = (-9, 14)$.

2 marks total
Multiply vectors
(–9, 14)

$\begin{pmatrix} -9 \\ 14 \end{pmatrix}$

(Total for Question 6 is 2 marks)

DO NOT WRITE IN THIS AREA